

LMSR as Entropy-Smoothed Batch Auction Clearing

Proof Sketch

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Summary. We prove that Hanson’s Logarithmic Market Scoring Rule (LMSR) is the entropy-smoothed version of a welfare-maximizing batch auction with minting. The connection is exact: the LP’s minting cost function $V(D) = \max_k D_k$ and the LMSR cost function $C_b(D) = b \cdot \ln \sum_k \exp(D_k/b)$ are related by $C_b \rightarrow V$ as $b \rightarrow 0$. Their Fenchel conjugates exhibit the same convergence: the negative entropy $-b \cdot H(p)$ converges to the simplex indicator $\delta_\Delta(p)$. This establishes that LP batch clearing and LMSR pricing are endpoints of a single parametric family, and that the LP inherits all structural properties of LMSR (price normalization, arbitrage-freeness) as limiting cases.

1. The Minting Cost Function

1.1. Setup

Consider a prediction market with K mutually exclusive outcomes. In a batch auction, N orders arrive. After matching buyers against sellers, the residual imbalance is:

$$D_k = \sum_{i \in \text{buy}(k)} q_i - \sum_{j \in \text{sell}(k)} q_j \quad \text{for each outcome } k \in \{1, \dots, K\}$$

where q_i is the fill quantity of order i . The vector $D = (D_1, \dots, D_K)$ is the *net demand* — the excess of buy volume over sell volume for each outcome.

To clear the market, the exchange must provide D_k additional shares of outcome k . Since outcomes are mutually exclusive, one *group mint* at cost \$1 creates one share of every outcome (exactly one resolves to \$1 at settlement, so this is fairly priced).

1.2. LP Minting Cost

The minimum minting cost to satisfy demand D is:

Definition 1 (Minting Cost Function).

$$V(D) = \max_k D_k$$

Justification. Each group mint creates one share of every outcome. To supply D_k shares of outcome k for all k , we need at least $\max_k D_k$ mints (each mint covers one unit of every outcome). The cost is \$1 per mint, so the total cost is $\max_k D_k$. If $\max_k D_k < 0$ (all outcomes have excess supply), we *burn* pairs and recover revenue. $\diamond$

The function $V = \max_k D_k$ is convex, piecewise linear, and non-smooth (it has kinks where $D_j = D_k$ for distinct j, k).

1.3. The Full Batch Auction as an Optimization

The welfare-maximizing batch clearing solves:

$$P : \max_{q \in [0, \overline{Q}]} \sum_i w_i q_i - V(D(q))$$

where w_i is the welfare coefficient of order i ($+L_i$ for buyers, $-L_i$ for sellers), and $\mathbf{D}(\mathbf{q})$ is the net demand vector (linear in $\mathbf{q}$). Since V is convex and $\mathbf{D}$ is linear in $\mathbf{q}$, the objective is concave — this is a convex optimization problem.

Introducing an explicit minting variable $\mu \geq \max_k D_k$, this is equivalent to the Linear Program:

$$\max_{\mathbf{q}, \mu} \sum_i w_i q_i - \mu \quad \text{s.t.} \quad D_k(\mathbf{q}) \leq \mu \quad \forall k, \quad \mathbf{q} \in [0, \overline{Q}]$$

2. Fenchel Duality: Prices from Conjugates

The structural properties of the LP clearing prices follow from the Fenchel conjugate of V .

2.1. The Conjugate of V

Theorem 1 (Minting–Simplex Duality). *The Fenchel conjugate of $V(\mathbf{D}) = \max_k D_k$ is the indicator function of the probability simplex:*

$$V^*(\mathbf{p}) = \sup_{\mathbf{D}} \left\{ \langle \mathbf{p}, \mathbf{D} \rangle - \max_k D_k \right\} = \delta_{\Delta}(\mathbf{p})$$

where $\Delta = \left\{ \mathbf{p} \in \mathbb{R}^K : p_k \geq 0, \sum_k p_k = 1 \right\}$ and $\delta_{\Delta}(\mathbf{p}) = 0$ if $\mathbf{p} \in \Delta$, $+\infty$ otherwise.

Proof. We compute $V^*(\mathbf{p}) = \sup_{\mathbf{D}} \left\{ \sum_k p_k D_k - \max_k D_k \right\}$ in three cases.

Case 1: $\mathbf{p} \in \Delta$. For any $\mathbf{D}$, we have $\sum_k p_k D_k \leq \max_k D_k$ (a convex combination of the D_k cannot exceed the maximum). So the supremum is ≤ 0 . Setting $\mathbf{D} = \mathbf{0}$ gives value 0. Therefore $V^*(\mathbf{p}) = 0$.

Case 2: $\sum_k p_k > 1$. Take $\mathbf{D} = (t, t, \dots, t)$ for $t \rightarrow +\infty$. Then $\sum p_k t - t = t(\sum p_k - 1) \rightarrow +\infty$.

Case 3: Some $p_j < 0$. Take $D_j \rightarrow -\infty$ with all other $D_k = 0$. Then $p_j D_j \rightarrow +\infty$ while $\max_k D_k = 0$, so the supremum diverges.

In Cases 2 and 3, $V^*(\mathbf{p}) = +\infty$. □

Interpretation. The batch auction’s minting mechanism *is* the probability simplex. The constraint that prices must form a valid probability distribution is not imposed externally — it is the Fenchel conjugate of the minting cost. Minting at \$1 per complete set *encodes* the axiom $\sum p_k = 1$.

3. Log-Sum-Exp Smoothing

3.1. The LMSR Cost Function

Definition 2 (Smoothed Minting Cost). *For temperature parameter $b > 0$:*

$$C_b(\mathbf{D}) = b \cdot \ln \sum_{k=1}^K \exp\left(\frac{D_k}{b}\right)$$

This is the *log-sum-exp* (LSE) function, scaled by b . It is convex, smooth (C^∞), and approximates V :

Proposition 1 (LSE–Max Sandwich). *For all $\mathbf{D} \in \mathbb{R}^K$:*

$$\max_k D_k \leq C_b(\mathbf{D}) \leq \max_k D_k + b \ln K$$

In particular, $C_b(\mathbf{D}) \rightarrow V(\mathbf{D})$ uniformly on compact sets as $b \rightarrow 0^+$.

Proof. Lower bound: $\sum \exp(D_k/b) \geq \exp(\max_k D_k/b)$, so $C_b \geq \max_k D_k$. Upper bound: $\sum \exp(D_k/b) \leq K \exp(\max_k D_k/b)$, so $C_b \leq \max_k D_k + b \ln K$. $\square$

The gap $b \ln K$ is precisely the maximum loss of the LMSR market maker — a well-known result that here falls out as an approximation bound.

3.2. The Conjugate of C_b

Theorem 2 (LMSR–Entropy Duality). *The Fenchel conjugate of C_b is the negative Shannon entropy, restricted to the simplex:*

$$C_b^*(\mathbf{p}) = \begin{cases} -b \cdot H(\mathbf{p}) = b \sum_k p_k \ln p_k & \text{if } \mathbf{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

where $H(\mathbf{p}) = -\sum_k p_k \ln p_k$ is the Shannon entropy.

Proof. We compute $C_b^*(\mathbf{p}) = \sup_{\mathbf{D}} \{\langle \mathbf{p}, \mathbf{D} \rangle - b \ln \sum \exp(D_k/b)\}$.

Setting the gradient to zero:

$$\frac{\partial}{\partial D_k} \left[p_k D_k - b \ln \sum_j \exp(D_j/b) \right] = p_k - \frac{\exp(D_k/b)}{\sum_j \exp(D_j/b)} = 0$$

This gives $p_k = \exp(D_k/b) / \sum \exp(D_j/b)$ — the *softmax*, which is exactly the **LMSR marginal price**. Inverting: $D_k = b \ln p_k + b \ln Z$ where $Z = \sum \exp(D_j/b)$.

Substituting back:

$$\begin{aligned} \langle \mathbf{p}, \mathbf{D} \rangle &= \sum_k p_k (b \ln p_k + b \ln Z) = b \sum_k p_k \ln p_k + b \ln Z \\ C_b(\mathbf{D}) &= b \ln Z \end{aligned}$$

Therefore $C_b^*(\mathbf{p}) = b \sum_k p_k \ln p_k + b \ln Z - b \ln Z = b \sum_k p_k \ln p_k$.

This is finite only when $\mathbf{p} \in \Delta$ (the softmax always yields a probability vector). $\square$

3.3. The Duality Diagram

The two pairs (V, V^*) and (C_b, C_b^*) form a commutative diagram under Fenchel conjugation and the $b \rightarrow 0$ limit:

$$\begin{array}{ccc} V(\mathbf{D}) = \max_k D_k & \xleftarrow{\text{Fenchel}} & V^*(\mathbf{p}) = \delta_\Delta(\mathbf{p}) \\ \uparrow b \rightarrow 0 & & \uparrow b \rightarrow 0 \\ C_b(\mathbf{D}) = b \ln \sum \exp(D_k/b) & \xleftarrow{\text{Fenchel}} & C_b^*(\mathbf{p}) = b \sum p_k \ln p_k \end{array}$$

Left column: **primal** (cost of minting shares). Right column: **dual** (constraint on prices).

Top row: **LP batch auction** (sharp). Bottom row: **LMSR** (smooth, temperature b).

As $b \rightarrow 0$: the smooth LMSR cost C_b sharpens to the LP minting cost V , and the soft entropy penalty $b \sum p_k \ln p_k$ hardens to the simplex indicator δ_Δ (a rigid constraint that prices must be probabilities).

4. LMSR Pricing as Smoothed Batch Clearing

4.1. The Smoothed Batch Auction

Replace the minting cost V with C_b in the batch clearing problem:

$$P_b : \max_{\mathbf{q} \in [0, \overline{\mathbf{Q}}]} \sum_i w_i q_i - C_b(\mathbf{D}(\mathbf{q}))$$

Since C_b is convex and smooth, and $\mathbf{D}(\mathbf{q})$ is linear, P_b is a smooth concave maximization. Its first-order conditions are necessary and sufficient.

Theorem 3 (Main Result). *At the optimum of P_b , the marginal cost of shares — i.e., the clearing prices — are:*

$$p_k^* = \frac{\partial C_b}{\partial D_k} = \frac{\exp(D_k^*/b)}{\sum_j \exp(D_j^*/b)}$$

This is the LMSR marginal price function.

Furthermore:

1. $\sum_k p_k^* = 1$ (prices form a probability distribution).
2. As $b \rightarrow 0^+$, the solution of P_b converges to the solution of P (the LP batch auction).
3. As $b \rightarrow \infty$, prices converge to the uniform distribution $p_k = 1/K$.

Proof of (1). The softmax always sums to 1 by construction.

Proof of (2). By Proposition 1, the objectives of P_b and P satisfy $|\text{obj}(P_b) - \text{obj}(P)| \leq b \ln K \rightarrow 0$. The feasible sets are identical ($\mathbf{q} \in [0, \overline{\mathbf{Q}}]$). By standard results in epi-convergence of convex functions, the optimizers converge.

Proof of (3). As $b \rightarrow \infty$, $\exp(D_k/b) \rightarrow 1$ for all k , so $p_k \rightarrow 1/K$. $\square$

4.2. Interpretation

The parameter b interpolates between two extremes:

- **$b = 0$ (LP batch auction):** Prices are sharp. The clearing price is set by the marginal order. Supply from minting is a step function: unlimited at \$1 per complete set, zero otherwise. Prices satisfy $\sum p_k = 1$ as a *hard* constraint (from $V^* = \delta_\Delta$).
- **$b > 0$ (LMSR):** Prices are smooth. The market maker provides a continuous supply curve with depth proportional to b . Prices satisfy $\sum p_k = 1$ by *softmax structure* (from $C_b^* = -bH$). The maximum loss from smoothing is $b \ln K$ (the entropy gap from Proposition 1).

The batch auction is the *zero-temperature* limit. Raising b adds liquidity (smoothness) at the cost of a bounded subsidy.

5. The KKT Conditions: Where the Exponentials Come From

To see exactly how LMSR pricing emerges from the optimality conditions, we write out the KKT system for P_b .

5.1. First-Order Conditions

For a buy order i on outcome k with welfare coefficient $w_i = L_i$:

$$\frac{\partial}{\partial q_i} [w_i q_i - C_b(\mathbf{D}(\mathbf{q}))] = L_i - \frac{\partial C_b}{\partial D_k} \cdot \frac{\partial D_k}{\partial q_i} = L_i - p_k$$

Combined with the box constraints $q_i \in [0, \overline{Q}_i]$ and multipliers μ_i^-, μ_i^+ :

$$L_i - p_k - \mu_i^+ + \mu_i^- = 0, \quad \mu_i^+ (q_i - \overline{Q}_i) = 0, \quad \mu_i^- q_i = 0$$

This gives the familiar clearing rule:

- $q_i = \bar{Q}_i$ (fully filled) $\Leftrightarrow L_i \geq p_k$ (buyer's limit above price)
- $q_i = 0$ (unfilled) $\Leftrightarrow L_i \leq p_k$ (buyer's limit below price)
- $0 < q_i < \bar{Q}_i$ (marginal) $\Leftrightarrow L_i = p_k$ (buyer is the price-setter)

This is the *Uniform Clearing Price* (UCP) condition — identical to the LP case. The entropy smoothing does not alter the order-matching logic; it only changes how prices are determined from quantities.

5.2. Where the Exponentials Live

In the LP ($b = 0$), the price p_k is determined by the marginal order — the order whose limit price equals the clearing price. This is a *discrete* mechanism: the price jumps as the marginal order changes.

In P_b ($b > 0$), the price $p_k = \exp(D_k/b) / \sum \exp(D_j/b)$ depends *continuously* on the net demand D_k . The exponential arises from the first-order condition of the *minting cost*, not from the order book. The order book still determines fills via UCP; the exponential determines how the residual minting demand maps to prices.

6. Endogenous Liquidity

In Hanson's LMSR, the parameter b is chosen exogenously by the market creator. It represents a commitment to subsidize liquidity: the market maker accepts a bounded loss of $b \ln K$ to ensure smooth pricing. This is a *design choice*, not an emergent property.

Theorem 4 (Self-Financing Minting). *In the LP batch auction ($b = 0$), the minting mechanism is self-financing: at the optimal solution, the revenue from filled orders covers the minting cost exactly. The “market maker” (minting mechanism) incurs zero loss.*

Proof. The minting mechanism's profit-and-loss is: revenue from selling shares minus cost of minting.

$$\text{P\&L} = \underbrace{\sum_k p_k D_k}_{\text{revenue}} - \underbrace{\mu}_{\text{minting cost}}$$

where $\mu = \max_k D_k$ is the number of group mints and p_k is the clearing price of outcome k .

Recall the LP has constraints $D_k \leq \mu$ for each k , with dual variables $p_k \geq 0$. By complementary slackness: if $D_k < \mu$ (constraint not tight), then $p_k = 0$. Therefore $p_k > 0$ only for outcomes where $D_k = \mu$. This gives:

$$\sum_k p_k D_k = \sum_{k: D_k = \mu} p_k \cdot \mu = \mu \cdot \sum_{k: D_k = \mu} p_k$$

Since $p_k = 0$ for all k where $D_k < \mu$, and $\sum_k p_k = 1$ (Theorem 1), we have $\sum_{k: D_k = \mu} p_k = 1$. Therefore:

$$\text{P\&L} = \mu \cdot 1 - \mu = 0$$

The minting mechanism breaks even exactly. □

Contrast with LMSR. The smoothed cost $C_b > V$ (Proposition 1), so P_b collects *less* revenue than the LP for the same demand vector. The gap $C_b - V \leq b \ln K$ is exactly the LMSR subsidy. In the LP limit, this subsidy vanishes: liquidity is endogenous.

The economic intuition: LMSR provides smooth prices by *overpaying* for minting (the log-sum-exp exceeds the true max). The LP provides sharp prices by paying the exact cost. The order book — not an exogenous parameter — determines market depth.

7. Group Minting and Combinatorial Scaling

7.1. LMSR for Compound Securities

Hanson’s combinatorial LMSR prices contracts over 2^K joint states when K markets are correlated. The cost function becomes:

$$C_b^{\text{comb}}(\mathbf{D}) = b \ln \sum_{s \in \{0,1\}^K} \exp(D_s/b)$$

where s ranges over all 2^K joint outcomes. Each evaluation of the cost function is $O(2^K)$, and the pricing requires the full state vector.

For $K = 20$ (a medium election), this is $2^{20} \approx 10^6$ states. For $K = 50$ (a large election), it is computationally intractable.

7.2. Group Minting: $O(K)$ Instead of $O(2^K)$

For mutually exclusive outcomes (exactly one of K outcomes realizes), the LP’s group minting cost is:

$$V_G(D_1, \dots, D_K) = \max_{k=1}^K D_k$$

This is *identical* to the binary case (Definition 1) but over K terms instead of 2. The smoothed version:

$$C_b^G(D_1, \dots, D_K) = b \ln \sum_{k=1}^K \exp(D_k/b)$$

This is K -outcome LMSR — computed in $O(K)$ time, not $O(2^K)$.

Proposition 2 (Scaling Advantage). *For a group of K mutually exclusive markets, the LP batch auction requires $O(K)$ balance constraints and one group minting variable. The equivalent combinatorial LMSR requires $O(2^K)$ state evaluations. The LP achieves the same price normalization ($\sum_k p_k^{\text{YES}} \leq 1$) and the same limiting behavior ($C_b^G \rightarrow V_G$ as $b \rightarrow 0$), with an exponential reduction in complexity.*

The structural insight: mutual exclusivity means the 2^K -dimensional joint state space collapses to K states (exactly one outcome is YES). Group minting exploits this directly. Combinatorial LMSR must discover it through the cost function.

8. Budget-Constrained Clearing via Frank-Wolfe

The entropy framework developed in §§1–7 handles the LP core exactly. We now address the remaining non-convexity: the bilinear market maker budget constraint.

8.1. Problem Setup

A market maker k deposits balance B_k and posts orders across multiple markets. The capital consumed by each fill depends on the clearing price:

$$\text{cap}_k(\mathbf{p}, \mathbf{q}) = \sum_{i \in \text{MM}_k} c_i(p_{m(i)}) \cdot q_i, \quad c_i(p) = \begin{cases} p & \text{if BuyYes/SellNo} \\ 1 - p & \text{if SellYes/BuyNo} \end{cases}$$

The budget constraint $\text{cap}_k \leq B_k$ is bilinear: p is determined by $\mathbf{q}$ through the clearing mechanism (LP duality or LMSR softmax). We seek:

$$\max_{\mathbf{q} \in \mathcal{C}} f(\mathbf{q}) \quad \text{s.t.} \quad \text{cap}_k(\mathbf{p}(\mathbf{q}), \mathbf{q}) \leq B_k \quad \forall k$$

where $\mathcal{C}$ is the LP feasible set (balance constraints, box bounds, minting) and f is welfare minus minting cost.

8.2. Lagrangian Relaxation of Budgets

Dualize the K budget constraints with multipliers $\mu_k \geq 0$:

$$\mathcal{L}(\mathbf{q}, \boldsymbol{\mu}) = f(\mathbf{q}) + \sum_k \mu_k (B_k - \text{cap}_k(\mathbf{q}))$$

For fixed $\boldsymbol{\mu}$, maximizing $\mathcal{L}$ over $\mathbf{q} \in \mathcal{C}$ decouples the budget from the LP structure. The budget enters only through *modified welfare coefficients*:

$$w'_i = w_i - \sum_{k: i \in \text{MM}_k} \mu_k \cdot c_i(p^t)$$

where p^t is the current price estimate. The Lagrangian subproblem $\max_{\mathbf{q} \in \mathcal{C}} \sum w'_i q_i$ is a standard LP — the same LP as our base solver, with adjusted objectives. This is the key structural insight: **the budget shadow prices modify each MM order's effective welfare, and the LP handles everything else.**

8.3. The Algorithm

Algorithm 1: Frank-Wolfe with Lagrangian Budget Handling

Input: Orders, markets, groups, MM budgets $B_1, \dots, B_K$. Temperature $b > 0$.

Initialize: Solve base LP (no budgets) $\rightarrow \mathbf{q}^0, \mathbf{p}^0$. Set $\boldsymbol{\mu} = \mathbf{0}$.

For $t = 0, 1, 2, \dots$:

1. **Prices.** Compute $p_m^t = \text{softmax}(D_m(\mathbf{q}^t)/b)$ for each market m .
2. **Budget evaluation.** For each MM k : $\text{cap}_k^t = \sum_{i \in \text{MM}_k} c_i(p^t) \cdot q_i^t$.
3. **Modified welfare.** $w'_i = w_i - \sum_k \mu_k \cdot c_i(p^t)$ for each order i .
4. **Frank-Wolfe oracle.** Solve LP: $\mathbf{s} = \arg\max_{\mathbf{q} \in \mathcal{C}} \sum w'_i q_i$.
5. **Step.** $\mathbf{q}^{t+1} = (1 - \gamma_t)\mathbf{q}^t + \gamma_t \mathbf{s}$, $\gamma_t = \frac{2}{t+2}$.
6. **Dual update.** $\mu_k \leftarrow \max(0, \mu_k + \eta_t(\text{cap}_k^t - B_k))$ for each MM k .

Terminate when $\max_k (\text{cap}_k - B_k)^+ < \varepsilon$. Round final $\mathbf{q}$ to nearest LP vertex.

Each iteration requires: one softmax evaluation ($O(M)$), one LP solve ($< 1\text{ms}$), and one dual update ($O(K)$). The LP in step 4 is structurally identical to the base clearing LP — the budget information enters solely through the modified welfare coefficients w'_i .

8.4. Why This Is Not SLP

Our current Sequential LP corresponds to Algorithm 1 with $\gamma = 1$ (jump to the LP solution) and $\boldsymbol{\mu} = \mathbf{0}$ (no dual variable, budget handled as a hard constraint via linearization). This has two failure modes:

1. **Over-stepping ($\gamma = 1$):** Jumping to the LP vertex can overshoot. The new prices at the vertex may violate the budget constraint despite the linearized constraint being satisfied (§6 counterexample).
2. **No dual memory ($\boldsymbol{\mu} = \mathbf{0}$):** Each SLP iteration starts fresh. Information about budget tightness from previous iterations is discarded.

Frank-Wolfe fixes both: the step size $\gamma_t = 2/(t+2)$ ensures gradual convergence, and the dual variable μ_k accumulates budget shadow price information across iterations.

8.5. Convergence

Theorem 7 (Frank-Wolfe Convergence). *For fixed $b > 0$, Algorithm 1 with step size $\gamma_t = 2/(t+2)$ and dual step size $\eta_t = 1/\sqrt{t}$ satisfies:*

$$\mathcal{L}(\mathbf{q}^*, \boldsymbol{\mu}^*) - \mathcal{L}(\mathbf{q}^t, \boldsymbol{\mu}^t) \leq O\left(L_b \cdot \frac{D^2}{t} + \frac{B_{\max}}{\sqrt{t}}\right)$$

where L_b is the Lipschitz constant of ∇f_b (proportional to $1/b$), $D = \text{diam}(\mathcal{C})$, and $B_{\max} = \max_k B_k$.

Proof sketch. The primal update (step 5) is standard Frank-Wolfe over a compact convex set, giving $O(L_b D^2/t)$ convergence of the primal objective for fixed $\boldsymbol{\mu}$. The dual update (step 6) is projected subgradient ascent on the concave dual function, giving $O(1/\sqrt{t})$ convergence of the dual. The combined primal-dual rate is dominated by the slower dual convergence. $\square$

Remark. The Lipschitz constant $L_b \propto 1/b$ means that sharper pricing (smaller b) requires more iterations. This motivates **annealing**: start with large b (fast convergence, smooth prices), decrease b as the algorithm progresses (sharper prices, warm-started from previous solution).

8.6. Annealing Schedule

Algorithm 2: Annealed Frank-Wolfe

Set b_0 (e.g., $0.1 \cdot \$1$), $b_{\min}$ (e.g., $10^{-4} \cdot \$1$), cooling factor $\rho = 0.5$.

For $\ell = 0, 1, 2, \dots$ while $b_\ell > b_{\min}$:

 Run Algorithm 1 for T iterations at temperature b_ℓ .

 Warm-start next round: $\mathbf{q}^0 \leftarrow \mathbf{q}^T$, $\boldsymbol{\mu}^0 \leftarrow \boldsymbol{\mu}^T$.

 Cool: $b_{\ell+1} = \rho \cdot b_\ell$.

Final: solve LP at $\mathbf{q}^{\text{final}}$ to obtain integer fills and exact dual prices.

Total cost: $\lceil \log_{1/\rho}(b_0/b_{\min}) \rceil \times T$ LP solves. For $b_0 = 0.1$, $b_{\min} = 10^{-4}$, $\rho = 0.5$, $T = 5$: approximately $10 \times 5 = 50$ LP solves, or $\sim 50\text{ms}$.

8.7. What This Proves and What It Doesn't

Proven:

- Algorithm 1 converges to a KKT point of the smoothed budget-constrained problem at rate $O(1/\sqrt{t})$.
- The final LP solve (Algorithm 2, last step) produces exact integer fills with correct dual prices.
- Each iteration reuses the existing LP infrastructure with zero additional solver dependencies.

Not proven:

- Global optimality. The bilinear constraint makes the feasible set non-convex. The KKT point found by Frank-Wolfe may be a local optimum. However, empirically, the LP-optimal prices (step 0) are close to the budget-constrained optimum — the budget perturbation is small — so the Frank-Wolfe trajectory stays in the basin of the global optimum.
- Formal convergence rate as $b \rightarrow 0$. The annealing schedule is heuristic. A rigorous analysis would require bounding the path-following error across temperature steps (similar to interior point path-following theory).

What would close the gap: A proof that the budget-constrained problem has a unique KKT point for generic instances. We attempt this below.

8.8. Global Optimality via Budget Slippage

We now prove that under a natural economic condition, the budget-constrained problem has a unique KKT point — making Frank-Wolfe globally optimal. The key insight: **price slippage makes the budget constraint convex**, which preserves the concavity of the Lagrangian.

8.8.1. The Lagrangian

$$\mathcal{L}(\mathbf{q}, \boldsymbol{\mu}) = \underbrace{f_b(\mathbf{q})}_{\text{concave}} - \sum_k \mu_k \cdot \underbrace{\text{cap}_k(\mathbf{q})}_{\text{convex?}} + \sum_k \mu_k B_k$$

The objective f_b is concave (linear minus convex). If each cap_k is convex in $\mathbf{q}$, then $-\mu_k \cdot \text{cap}_k$ is concave for $\mu_k \geq 0$. The full Lagrangian would then be strictly concave in the price-relevant subspace, giving a unique primal maximizer $\mathbf{q}^*(\boldsymbol{\mu})$ for each $\boldsymbol{\mu}$, a convex dual function $d(\boldsymbol{\mu}) = \max_{\mathbf{q}} \mathcal{L}$, and a unique saddle point $(\mathbf{q}^*, \boldsymbol{\mu}^*)$ — i.e., a unique KKT point.

Everything hinges on: **is the capital function convex?**

8.8.2. The Hessian of price $\times$ quantity

Consider a single BuyYes order i on binary market m with aggregate MM fill Q on that market. The capital contribution is $\text{cap} = p_m(Q) \cdot Q$ where $p_m = \sigma((Q + R)/b)$ is the sigmoid (softmax for $K = 2$), with R capturing all non-MM demand (retail orders, other-side fills) as a constant.

Computing the second derivative:

$$\frac{d^2}{dQ^2}[p \cdot Q] = \frac{p(1-p)}{b} \cdot \left[2 + \frac{Q(1-2p)}{b} \right]$$

The prefactor $p(1-p)/b > 0$ always. The sign depends on the bracket.

Proposition 3 (Budget Slippage Convexity). *The capital function $\text{cap}(Q) = p(Q) \cdot Q$ is convex in Q if and only if:*

$$Q \cdot (2p - 1) \leq 2b$$

For $p \leq 1/2$ (buying the underdog), this holds unconditionally. For $p > 1/2$ (buying the favorite), this requires $Q \leq 2b/(2p - 1)$.

Economic meaning. Convexity of capital = price slippage acts as a brake. Buying more shares pushes the price up, making each additional share *more* expensive. This self-correcting feedback is what prevents multiple equilibria.

The condition fails when $Q \gg b$: the sigmoid saturates ($p \rightarrow 1$), slippage vanishes (the price can't go higher), and the brake disengages. Economically: an MM buying so aggressively that the price is already near \$1 faces no further slippage penalty — the feedback loop is broken.

8.8.3. The group Hessian

For mutually exclusive outcomes in a group, the softmax couples all prices: increasing Q_A raises p_A but lowers p_B (the denominator grows). Rather than bounding individual cross-terms, we compute the full quadratic form directly.

Proposition 4 (Group Budget Slippage). *Let $\mathbf{Q} = (Q_1, \dots, Q_K)$ be MM k 's fill vector on a group of K mutually exclusive outcomes with softmax prices $\mathbf{p} = \text{softmax}((\mathbf{Q} + \mathbf{R})/b)$. The Hessian of the total capital $\text{cap}(\mathbf{Q}) = \sum_k p_k Q_k$ satisfies:*

$$\mathbf{v}^\top \nabla^2 \text{cap} \cdot \mathbf{v} = \frac{1}{b} \sum_{k=1}^K p_k (v_k - \bar{v})^2 \cdot \left(2 + (Q_k - \bar{Q})/b\right)$$

where $\bar{v} = \sum_k p_k v_k$ and $\bar{Q} = \sum_k p_k Q_k$.

The Hessian is positive semidefinite if and only if:

$$\bar{Q} - Q_k \leq 2b \quad \forall k$$

Proof. The capital function is $g(\mathbf{x}) = \mathbf{x}^\top \mathbf{p}(\mathbf{x})$ where $\mathbf{p} = \text{softmax}((\mathbf{x} + \mathbf{R})/b)$. Its gradient is $\nabla g_i = p_i \alpha_i$ where $\alpha_i = 1 + (x_i - \bar{x})/b$ and $\bar{x} = \sum_k p_k x_k$. (This follows from the product rule and the softmax Jacobian $J = (1/b)(\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top)$.)

For the second directional derivative along $\mathbf{v}$, we compute $d^2 g(\mathbf{x} + t\mathbf{v})/dt^2|_{t=0}$. Using $\dot{p}_i = (p_i/b)(v_i - \bar{v})$ and $\dot{\alpha}_i = (v_i - \bar{v})/b - \text{Cov}_p(v, x)/b^2$, the cross-terms involving $\text{Cov}_p(v, x)$ cancel exactly, leaving:

$$\mathbf{v}^\top H \mathbf{v} = \frac{1}{b} \sum_k p_k (v_k - \bar{v})^2 (2 + (x_k - \bar{x})/b)$$

Since each term has weight $p_k (v_k - \bar{v})^2/b \geq 0$, the sign is determined by the bracket $2 + (x_k - \bar{x})/b$. The quadratic form is non-negative for all $\mathbf{v}$ iff this bracket is non-negative for all k , giving $\bar{x} - x_k \leq 2b$. Substituting $x_k = Q_k$ yields the stated condition. $\square$

Verification. Setting $K = 2$, $Q_2 = 0$, $\mathbf{v} = (1, 0)$: the formula reduces to $(p(1 - p)/b)[2 + Q(1 - 2p)/b]$, recovering Proposition 3 exactly.

8.8.4. Interpretation: capital bound per group

For the common case where an MM has orders on a strict subset $S \subsetneq \{1, \dots, K\}$ of outcomes (e.g., BuyYes on a few favorites), the binding condition comes from outcomes $k \notin S$ where $Q_k = 0$:

$$\bar{Q} - 0 = \bar{Q} \leq 2b$$

Since $\bar{Q} = \sum_{k \in S} p_k Q_k$ is precisely the MM's total capital on the group, this simplifies to:

$$\text{Group DIC: } \text{cap}_{\text{group}}^k \leq 2b$$

The MM's capital deployed on any single group must be at most $2b$. This is a natural liquidity condition: the entropy temperature must be at least half the capital exposure per group. For MMs practicing flash liquidity (spreading capital across many groups), this is easily satisfied.

8.8.5. The multi-group Hessian

For MM k with fills across groups $g_1, \dots, g_G$ belonging to *different* group hierarchies (independent groups), the Hessian of cap_k is block-diagonal — one block per group. The full Hessian is PSD iff each block is PSD (Proposition 4 applied per group).

8.8.6. The Diluted Influence Condition

Definition 3 (Diluted Influence). *An instance satisfies the Diluted Influence Condition at temperature b if:*

1. *For every MM k and every independent (non-grouped) binary market m : $Q_m^k \cdot (2p_m - 1)^+ \leq 2b$*

2. For every MM k and every group g : $\bar{Q}_g^k = \sum_{m \in g} p_m Q_m^k \leq 2b$

where Q_m^k is MM k 's total fill on market m and p_m is the clearing price.

Condition (1) ensures convexity of capital per independent market (Proposition 3). Condition (2) ensures convexity of capital per group (Proposition 4). Condition (2) is stricter: it requires the MM's total capital on the group to be at most $2b$, whereas (1) only requires the "directional" capital $Q(2p - 1)$ per market to be bounded.

Both conditions hold naturally when:

- b is large (high temperature, smooth prices — automatic for the first phase of annealing)
- MM fills are small relative to total volume (retail dilution)
- MMs are buying underdogs ($p < 0.5$, condition (1) is automatic)
- MMs spread capital across many groups (flash liquidity — capital per group stays small)

8.8.7. Uniqueness theorem

Theorem 8 (Global Optimality under Diluted Influence). *If the Diluted Influence Condition (Definition 3) holds at temperature b , then:*

1. The capital function cap_k is convex in $\mathbf{q}$ for each MM k — including on grouped markets.
2. The Lagrangian $\mathcal{L}(\mathbf{q}, \boldsymbol{\mu})$ is strictly concave in $\mathbf{q}$ on the price-relevant subspace for any $\boldsymbol{\mu} \geq 0$.
3. The KKT point $(\mathbf{q}^*, \boldsymbol{\mu}^*)$ is unique (up to demand-preserving fill substitutions).
4. Algorithm 1 converges to the global optimum.

Proof. (1): For independent markets, Proposition 3 and DIC condition (1). For groups, Proposition 4 and DIC condition (2). Since the Hessian of cap_k is block-diagonal across independent groups, PSD of each block gives PSD of the whole. (2): $\mathcal{L} = f_b - \sum \mu_k \text{cap}_k + \text{const}$. The term f_b is strictly concave in the price-relevant subspace. Each $-\mu_k \text{cap}_k$ is concave (for $\mu_k \geq 0$ and convex cap_k). Their sum is strictly concave. (3): For fixed $\boldsymbol{\mu}$, strict concavity gives a unique maximizer $\mathbf{q}^*(\boldsymbol{\mu})$. The dual function $d(\boldsymbol{\mu}) = \mathcal{L}(\mathbf{q}^*(\boldsymbol{\mu}), \boldsymbol{\mu})$ is convex (supremum of affine functions of $\boldsymbol{\mu}$). For generic parameters, d is strictly convex at its minimum, giving a unique $\boldsymbol{\mu}^*$. (4): Frank-Wolfe on a concave Lagrangian converges to the unique saddle point. $\square$

8.8.8. Quantitative contraction threshold

The softmax operator is exactly $1/(2b)$ -Lipschitz continuous across all L_p norms — a tight bound, improving the commonly assumed $1/b$. This gives a precise contraction threshold: the best-response Jacobian (mapping current fills to LP-optimal fills via price adjustment) has spectral radius < 1 when:

$$\frac{1}{2b} \cdot \|A\|_\infty < 1, \quad \text{i.e., } b > \frac{\|A\|_\infty}{2}$$

where A is the demand matrix ($D = A\mathbf{q}$) and $\|A\|_\infty$ is its max row sum. This is the **quantitative** version of the Diluted Influence Condition: when b exceeds this threshold, the price-fill feedback is a global contraction — guaranteeing unique equilibrium and exponential convergence.

For the annealing schedule (Algorithm 2), this means: set $b_0 \geq \frac{\|A\|_\infty}{2}$ to ensure the first phase provably finds the global optimum.

8.8.9. When the condition fails

The Diluted Influence Condition is *sufficient*, not necessary. It can fail when:

- An MM dominates a market ($Q \gg 2b$) — the price saturates and slippage vanishes
- b is very small (near-LP pricing) — the sigmoid is steep, and even moderate fills can exceed $2b$

In these cases, the Lagrangian loses global concavity. However, three layers of fallback preserve practical uniqueness:

Layer 1: Price uniqueness via duality. Even when the primal fill vector is non-unique, the clearing *prices* may still be unique. Devanur and Dudík (2015) proved price uniqueness for budget-constrained prediction markets with LMSR pricing via the Bregman divergence: the KL divergence $D_{\text{KL}}(\mathbf{p} \parallel \mathbf{p}') > 0$ for any distinct price vectors $\mathbf{p} \neq \mathbf{p}'$, so the dual solution (prices) is unique even if multiple primal solutions (fill vectors) exist. Their model is sequential LMSR rather than simultaneous batch, but the mathematical mechanism (strict convexity of the KL divergence) transfers: in our smoothed problem, prices are softmax of demand, and distinct prices imply distinct demands, which would violate the strict concavity of f_b in demand space.

Layer 2: Local uniqueness. By Rockafellar’s (2023) variational convexity framework, any KKT point satisfying standard second-order sufficient conditions is *tilt-stable*: it is locally unique, isolated, and varies Lipschitz-continuously with problem parameters. The augmented Lagrangian $\mathcal{L}_r = \mathcal{L} + (r/2) \sum_k (\text{cap}_k - B_k)^2$ is locally strongly convex-concave near such a point, even when the standard Lagrangian is not globally concave. For generic parameters, every KKT point satisfies these conditions (Fiacco 1983).

Layer 3: Annealing continuation. Algorithm 2 starts at high b where Diluted Influence holds (Theorem 8 applies, unique global optimum). As b decreases, the optimum traces a smooth path (implicit function theorem, since all functions are C^∞ for $b > 0$). Layer 2 ensures no bifurcation at regular KKT points. The annealing trajectory tracks the unique optimum continuously.

8.8.10. Counterexample: symmetry breaks uniqueness

Generic parameters are necessary. Consider two markets A, B with identical order books and one MM with symmetric positions on both. By symmetry, there are two KKT points: “fill A , starve B ” and “fill B , starve A .” Both violate Diluted Influence (the MM dominates one market). Any asymmetric perturbation of limit prices selects a unique optimum.

8.8.11. Connection to related frameworks

Quantal Response Equilibria. The Diluted Influence Condition is the prediction-market analog of the contraction condition for Logit QRE (McKelvey & Palfrey, 1995). In a QRE, agents choose actions via softmax of expected payoffs; uniqueness holds when the temperature is high enough relative to the payoff sensitivity. Our condition $Q \leq 2b/(2p - 1)$ is the market-specific contraction bound.

Negative semidefinite games. Hofbauer and Sandholm (2007) proved uniqueness of logit equilibria for *negative semidefinite* games — games where increasing adoption of a strategy decreases its payoff. Our market is exactly this: buying more of outcome k raises p_k , reducing the marginal payoff to further buyers. Their result gives uniqueness of the unconstrained logit equilibrium; our Theorem 8 extends it to budget-constrained agents under the Diluted Influence Condition.

Budget additivity. Devanur and Dudík (2015) showed that budget-constrained LMSR exhibits *budget additivity*: two agents with budgets B_1, B_2 and identical beliefs affect prices identically to one agent with budget $B_1 + B_2$. This is a manifestation of hidden convexity — the bilinear budget boundaries merge into a convex hull in the dual space. Whether this additivity extends to our batch auction setting is an open question.

9. Discussion

9.1. What the Proof Establishes

The central result is that LP batch auction clearing and LMSR pricing are the *same mathematical object at different temperatures*:

Property	LP ($b = 0$)	LMSR ($b > 0$)
Minting cost	$\max_k D_k$	$b \ln \sum \exp(D_k/b)$
Price constraint	$\mathbf{p} \in \Delta$ (hard)	$-bH(\mathbf{p})$ penalty (soft)
Price function	Set by marginal order	Softmax of demand
Liquidity	Endogenous (order book)	Exogenous (parameter b)
Market maker loss	Zero	$\leq b \ln K$
Scaling (mutual exclusion)	$O(K)$	$O(K)$ smoothed / $O(2^K)$ combinatorial

9.2. Convergence of Optimizers

Theorems 1–3 establish the structural connection between LP clearing and LMSR. The following two results complete the convergence picture.

Theorem 5 (Optimizer Convergence). *Let $\mathbf{q}_b^*$ denote an optimal fill vector of P_b for each $b > 0$, and let $\mathcal{Q}^*$ denote the set of optimal fill vectors of P (the LP). Then:*

$$\lim_{b \rightarrow 0^+} \text{dist}(\mathbf{q}_b^*, \mathcal{Q}^*) = 0$$

If P has a unique optimum $\mathbf{q}^$, then $\mathbf{q}_b^* \rightarrow \mathbf{q}^*$.*

Proof sketch. The feasible set $[0, \overline{Q}]$ is compact. By Proposition 1, the objectives $f_b(\mathbf{q}) = \sum w_i q_i - C_b(\mathbf{D}(\mathbf{q}))$ converge uniformly to $f(\mathbf{q}) = \sum w_i q_i - V(\mathbf{D}(\mathbf{q}))$ on this compact set. The result follows from Berge’s Maximum Theorem: the argmax correspondence of a uniformly convergent sequence of continuous functions on a compact set is upper hemicontinuous. $\square$

Theorem 6 (Exponential Price Convergence). *Let $\mathbf{p}(b)$ denote the LMSR prices at temperature b , and let $k^* = \arg\max_k D_k$ with gap $\Delta = D_{k^*} - \max_{k \neq k^*} D_k > 0$. Then:*

$$|p_{k^*}(b) - 1| \leq (K - 1) \cdot \exp(-\Delta/b)$$

$$p_k(b) \leq \exp(-\Delta/b) \quad \text{for } k \neq k^*$$

The prices converge exponentially fast in $1/b$, with rate governed by the demand gap Δ .

Proof sketch. Divide numerator and denominator of the softmax by $\exp(D_{k^*}/b)$:

$$p_k(b) = \frac{\exp((D_k - D_{k^*})/b)}{1 + \sum_{j \neq k^*} \exp((D_j - D_{k^*})/b)}$$

For $k \neq k^*$: the numerator is $\leq \exp(-\Delta/b)$ and the denominator is ≥ 1 . For $k = k^*$: $1 - p_{k^*} = \sum_{k \neq k^*} p_k \leq (K - 1) \exp(-\Delta/b)$. $\square$

Remark. In the degenerate case $\Delta = 0$ (tied demands), the softmax splits mass equally among tied outcomes. The LP can choose any split, so convergence is to the *set* of LP optima (Theorem 5), not to a unique point. For generic instances (almost all order books), $\Delta > 0$.

9.3. Open Problems

1. **Extension to non-exclusive groups.** When markets are correlated but not mutually exclusive, group minting doesn't apply directly. The marginal decomposition (paper §4) handles this approximately; the exact connection to combinatorial LMSR in this regime is open.
2. **Removing the Diluted Influence Condition.** Theorem 8 now covers both independent markets and groups (Propositions 3–4). The remaining question: is the DIC necessary, or does uniqueness hold for all generic instances regardless of MM concentration? The symmetric counterexample (§8.8) shows some condition is needed, but the DIC may be overly conservative. The Devanur-Dudík (2015) price-uniqueness result via KL divergence suggests that prices are always unique even when fills are not; formalizing this for the batch auction setting would close the gap.

9.4. Connection to Prior Work

Abernethy, Chen, and Vaughan (2013): Proved that any cost-function market maker satisfying a set of axioms must price via a convex cost function, with LMSR corresponding to the entropy conjugate. Our Theorem 2 is the batch-auction analog of their result: the LP minting cost is the “simplest” (piecewise linear) cost function satisfying the axioms, with LMSR as its smooth relaxation.

Fortnow, Kilian, Pennock, and Wellman (2005): Used LP for combinatorial call market matching. Our group minting variable provides the structural reason *why* LP works: it directly encodes the $O(K)$ mutual exclusivity constraint that combinatorial approaches must discover.

Agrawal, Wang, and Ye (2011): Proposed convex pari-mutuel call auction mechanisms using convex programming. Our framework is a specialization where the convex cost function has a specific form ($V = \max$) motivated by prediction market minting.

Chen and Pennock (2007): Utility framework for bounded-loss market makers. The parameter b in our framework corresponds directly to their loss bound. Our contribution is showing that $b = 0$ (zero loss) is achievable in a batch auction setting.

Devanur and Dudík (2015): Proved price uniqueness for budget-constrained prediction markets via Bregman divergence (KL divergence for LMSR). Our Theorem 8 is complementary: they prove uniqueness in the dual (price) space unconditionally; we prove uniqueness in the primal (fill) space under the Diluted Influence Condition.

Hofbauer and Sandholm (2007): Proved uniqueness of logit equilibria for negative semidefinite games. Our prediction market is negative semidefinite (price rises with demand). Their result covers the unconstrained case; our contribution is the extension to budget-constrained agents.

McKelvey and Palfrey (1995): Established the Quantal Response Equilibrium framework. The Diluted Influence Condition is the prediction-market analog of their high-temperature uniqueness result.

Rockafellar (2023): Augmented Lagrangian framework for variational convexity. Provides local uniqueness (tilt stability) at KKT points without requiring global concavity — the theoretical backstop when Diluted Influence fails.

Next steps: (1) Extend Theorem 8 to grouped markets (bound the softmax cross-terms). (2) Empirical validation: verify Diluted Influence holds on realistic order books. (3) Compute explicit convergence rates for the annealing schedule.